

**MBA RAPID FIRE**

# **CAT 2025 - FINAL LAP**

**FOR MBA ASPIRANTS**

**Lecture - 03**

**Arithmetic - 3**

**By- RAHUL BATHLA SIR**



**CAT 99.95 %ile**

# **TOPICS** TO BE COVERED

1

**CHEAT SHEETS**

2

**MIXED BAG QUESTIONS**

# CHEAT SHEETS

A horizontal graphic of a pencil with a blue light effect emanating from both ends, passing behind the text.

WORK RATE TIME

□ A finish work in 6 Days

□ B finish work in 4 Days

□ Work = 12

□ Rate (A) = 2

□ Rate (B) = 3

|      | A + B |
|------|-------|
| Work | 12    |
| Rate | 5     |
| Time | 2.4   |

□ A work for 1 Day , B joins A from Day 2. Cost (1000 Rs)

|      | A | A + B |
|------|---|-------|
| Work | 2 | 10    |
| Rate | 2 | 5     |
| Time | 1 | 2     |

$$W_B = 2(3) = 6$$

$$W_A = 12 - 6 = 6$$

▪ Cost (A) = 500

▪ Rate (B) = 500

□ A work on 1 Day , B work on Day 2 (Repeat pattern)

|      | A | B |
|------|---|---|
| Work | 2 | 3 |
| Rate | 2 | 3 |
| Time | 1 | 1 |

1 Cycle - 2 Days - 5 Work

2 Cycle - 4 Days 10 Work

+ 1 DAY → 5 DAYS 12 Work

- ❑ If 6 men and 8 boys can do a piece of work in 10 days
- ❑ While 26 men and 48 boys can do the same in 2 days

$$(6M + 8B) \times 10 = (26M + 48B) \times 2$$

$$W = (6+8) \times 10 = 20 \times 10 = 200$$

$$M = 2$$

$$B = 1$$

$$30M + 40B = 26M + 48B$$

$$4M = 8B$$

$$M = 2B$$

$$\frac{M}{B} = \frac{2}{1}$$

## WORK RATE TIME - 3 - (QUESTION - 1)



- ❖ A and B together can finish the work in half of the time than the time B takes alone to finish the work working at it's 80% efficiency.
- ❖ If A takes 4 more days than B to finish the work.
- ❖ Find the time A will take to finish the work with B if B work's with his 60% efficiency.

$$T_{A+B} = \frac{1}{2} T_{B'}$$

$$A+B = 2 B'$$

$$A+B = 2 (0.8B)$$

$$A+B = 1.6B$$

$$A = 0.6B$$

$$A = \frac{3}{5} B$$

$$5A = 3B$$

$$T_a = x + 4$$

$$T_b = x$$

$$W = 5(6) = 30$$

$$x = 6$$

|      | A         | B    |
|------|-----------|------|
| Work | $3x + 12$ | $5x$ |
| Rate | 3         | 5    |
| Time | $x + 4$   | $x$  |

|      | A + B'      |
|------|-------------|
| Work | 30          |
| Rate | $3 + 3 = 6$ |
| Time | 5           |

Q.

## QUESTION - 2

$$a > 10$$

$$a + b = 20$$

$$10 \uparrow \quad 10 \downarrow$$



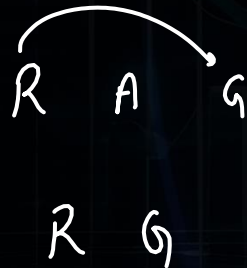
#Q :2 Rahul, Akshita and Gurmeet, working together, would have taken more than 7 days to finish a job. On the other hand, Rahul and Gurmeet, working together would have taken less than 15 days to finish the job. However, they all worked together for 6 days, followed by Akshita, who worked alone for 3 more days to finish the job. If Akshita had worked alone on the job then the number of days she would have taken to finish the job, cannot be

$$\frac{105}{5} \quad \frac{105}{7}$$

$$21 > T > 15$$

$$5 < A < 7$$

**A** 17  
 $R + G = C$



$$C + A < 15$$

$$C > 7$$

**B** 21  
T  
7<sup>+</sup> W 105 R 15<sup>-</sup>  
15<sup>-</sup> 105 7<sup>+</sup>

**C** 16

**D** 20

$$6(R + A + G) + 3A = 105$$

$$2(R + A + G) + A = 35$$

$$2(C + A) + A = 35$$

$$30 \downarrow \quad 5 \uparrow$$

$$\underbrace{2C}_{14 \uparrow} + \underbrace{3A}_{21 \downarrow} = 35$$



## UNIQUE CYCLIC PATTERNS



$$A=1 \quad B=2$$

$A, B, AB, A, B, AB, \dots$

$\underbrace{\quad}_1 \quad \underbrace{\quad}_2 \quad \underbrace{\quad}_3$

1 Cycle  $\rightarrow$  3 Days  $\rightarrow$  6 Work

A works every day  $A=1$   
 $B=2$   
 $C=3$

B works every second day

C works every 3rd day

$$\left. \begin{array}{l} \text{fr } A \rightarrow 1 \\ \text{fr } B \rightarrow 2 \\ \text{fr } C \rightarrow 3 \end{array} \right\} \text{LCM} = \underline{\underline{6}}$$

|       |       |          |
|-------|-------|----------|
| DAY 1 | A     | 1        |
| DAY 2 | A B   | 3        |
| DAY 3 | A C   | 4        |
| DAY 4 | A B   | 3        |
| DAY 5 | A     | 1        |
| DAY 6 | A B C | 6        |
|       |       | <hr/> 18 |

1 Cycle 18 Work 6 Days



# CHEAT SHEETS

DTS

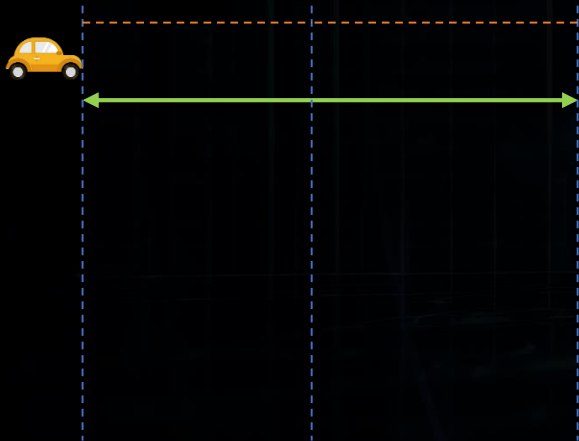


$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

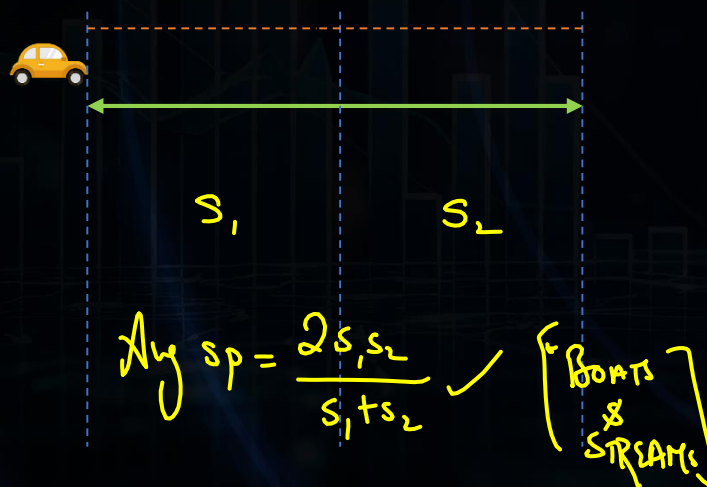
$$\text{Avg Sp} = \text{AM} = \frac{s_1 + s_2}{2}$$

Time Constant



$$\text{Avg Sp} = \text{HM} = \frac{n}{\frac{1}{s_1} + \frac{1}{s_2} + \dots + \frac{1}{s_n}}$$

Distance Constant



- If Speed becomes double then time becomes  $\frac{1}{2}$

$$\boxed{\frac{s_1}{s_2} = \frac{T_2}{T_1}}$$

- If Time becomes  $\frac{3}{2}$  times then speed becomes  $\frac{2}{3}$

- ❖ A beats B by 50 meters in a Race and B beats C by 80 mtrs in same race. If A beats C by 140mtrs in same race. Find the length of the race.

$$\frac{A}{B} = \frac{x}{x - 50}$$

$$\frac{B}{C} = \frac{x}{x - 80}$$

$$\frac{A}{C} = \frac{x}{x - 140}$$

$$\boxed{\frac{A}{C} = \frac{A}{B} \times \frac{B}{C}}$$

$$L = \frac{ab}{a+b-c}$$

$$\begin{array}{lcl} A \rightarrow B & a \\ B \rightarrow C & b \\ A \rightarrow C & c \end{array}$$

- ❖ A beats B by 50 meters or 5 sec in a Race of 100 m.

$$\frac{A}{B} = \frac{100}{50} = \frac{Va}{Vb}$$

$\underbrace{\hspace{10em}}_{100 - 50}$

$$S_B = \frac{50}{s} = 10 \text{ m/s}$$

# CHEAT SHEET - 3



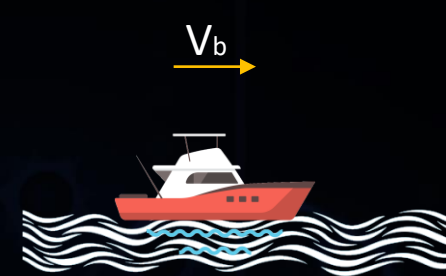
$50 \text{ km}$   
 $9:00$   $S = 50 (T)$   $10:00 \text{ pm}$   
 $S = 40 (T + \frac{1}{4})$   $10:15 \text{ pm}$   
 $50T = 40T + 10$   
 $10T = 10$   $T = 1$

$10:00 \text{ pm}$   $S$   $T$   
 $10:15 \text{ pm}$   $(S + 10) (T - \frac{1}{4})$

- A boat goes 8 Km upstream and 6 Km Downstream in 12 hours.

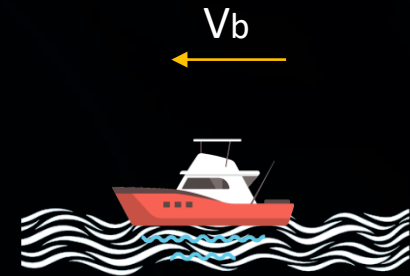
$$\frac{8}{V_b - V_w} + \frac{6}{V_b + V_w} = 12$$

## DOWNSTREAM



$$V_d = V_b + V_w$$

$$V_b = \frac{V_d + V_u}{2}$$

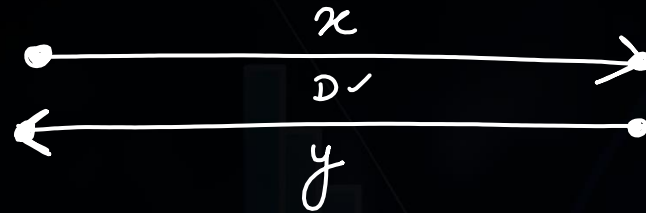


$$V_u = V_b - V_w$$

$$V_w = \frac{V_d - V_u}{2}$$

$$\frac{V_d}{V_u} = \frac{V_b + V_w}{V_b - V_w} = \frac{T_u}{T_d}$$

# AVERAGE SPEED IN UPSTREAM - DOWNSTREAM



$$D = \frac{2xy}{x+y} (T)$$

$$D = \frac{(v_B + v_w)(v_B - v_w)}{v_B + v_w + v_B - v_w} \times (T)$$

$$D = \frac{v_B^2 - v_w^2}{2v_B} (T)$$

Labels in the diagram:

- $\frac{v_B^2 - v_w^2}{2v_B}$  is labeled as  $\frac{v_B^2 - v_w^2}{2v_B}$ .
- $(T)$  is labeled as **TOTAL TIME (AANA + JAANA)**.
- $D$  is labeled as **Dist b/n A & B**.

#Q. A man travels by a motor boat down a river to his office and back. With the speed of the river unchanged, if he doubles the speed of his motor boat, then his total travel time gets reduced by 75%<sup>25%</sup>. The ratio of the original speed of the motor boat to the speed of the river is:

**A**  $\sqrt{6} : \sqrt{2}$

**B**  $\sqrt{7} : 2$  ✓

**C**  $2\sqrt{5} : 3$

**D**  $3 : 2$

Initial

$V_B$

$V_w$

✓ D

4

Final

$2V_B$

$V_w$

D ✓

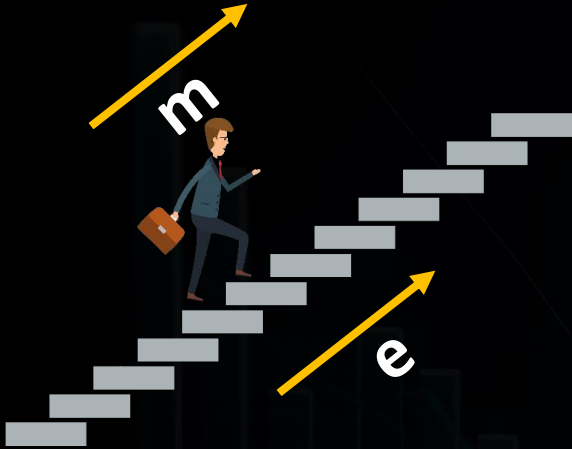
1

$$\left[ \frac{V_B^2 - V_w^2}{2} \right]_4 = \left[ \frac{4V_B^2 - V_w^2}{2} \right]_1$$

$$8V_B^2 - 8V_w^2 = 4V_B^2 - V_w^2$$

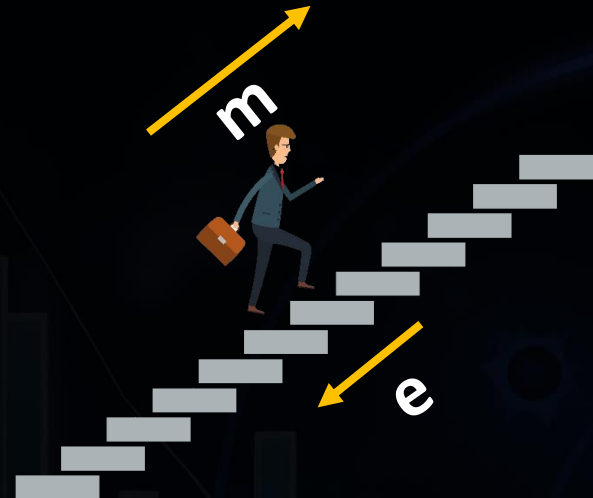
$$4V_B^2 = 7V_w^2$$

$$\frac{V_B^2}{V_w^2} = \frac{7}{4}$$



$$n = (m + e)t$$

$$n = \underbrace{mt}_{\text{steps by men}} + \underbrace{et}_{\text{steps by esc.}}$$



$$n = (m - e)t$$

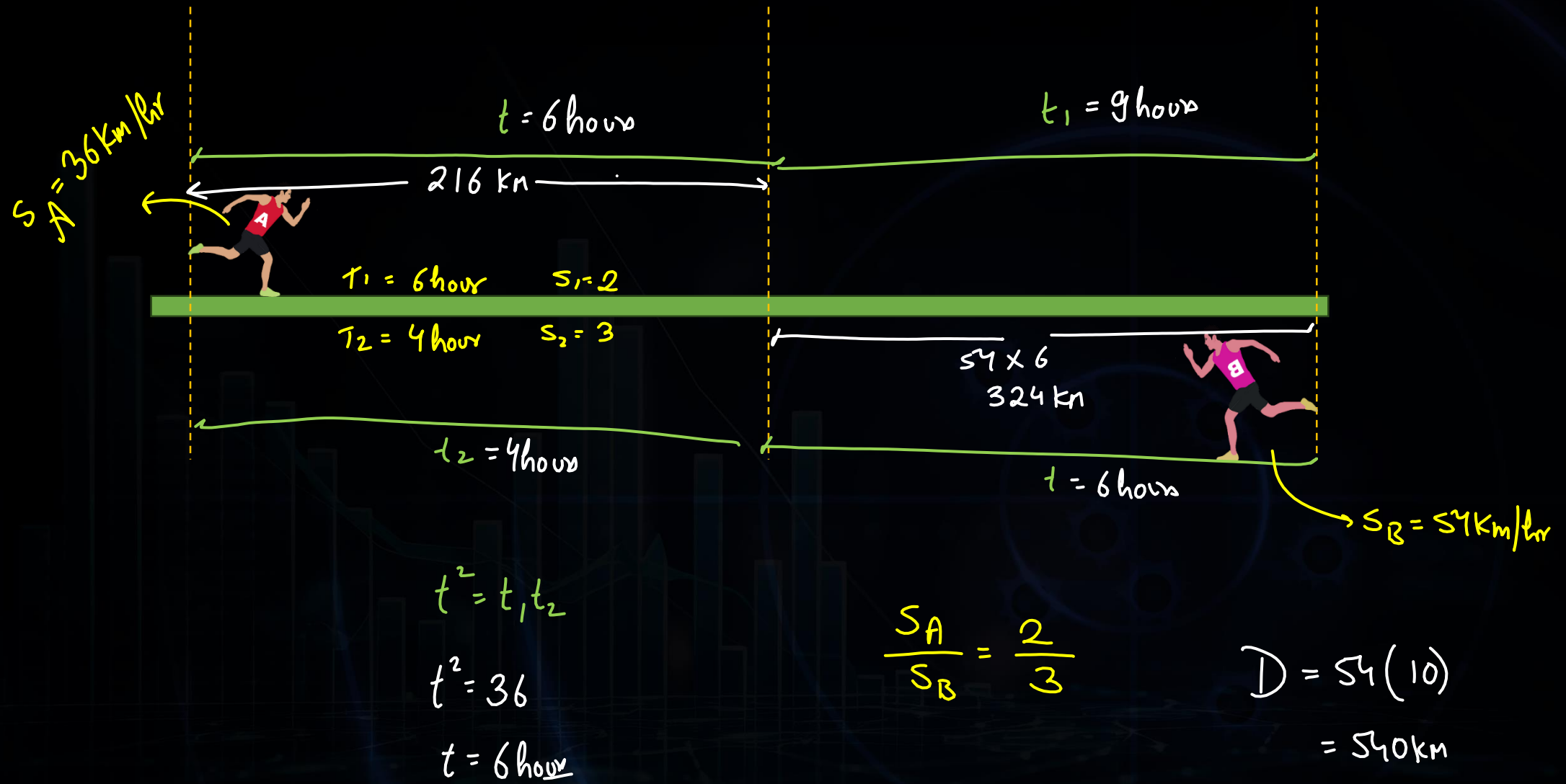
$$n = \underbrace{mt}_{\text{steps by men}} - \underbrace{et}_{\text{steps by esc.}}$$

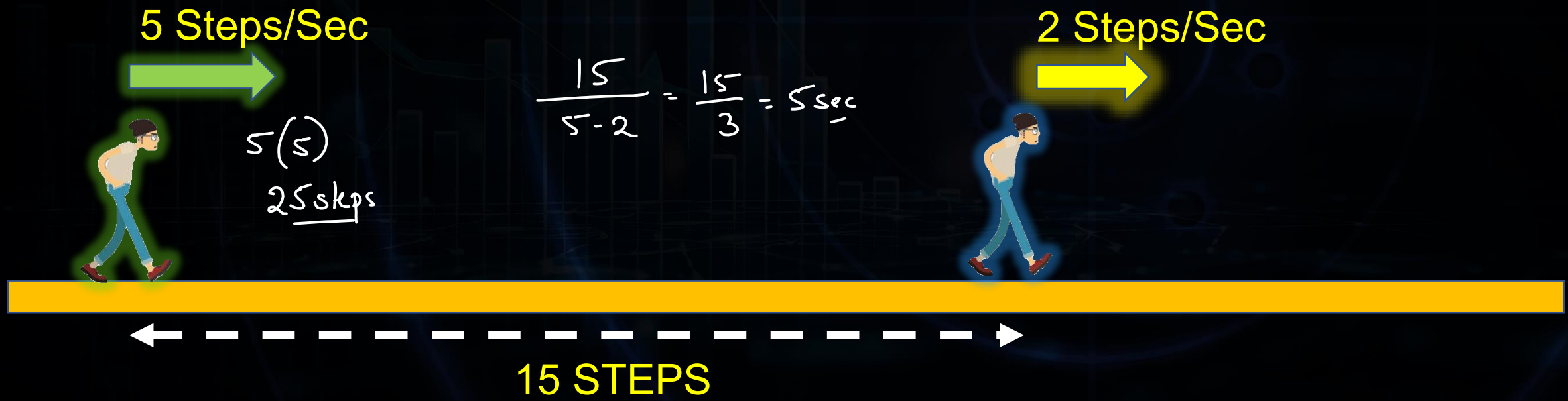
❖ { Kamal takes 20 steps and 10 sec to reach from 1<sup>st</sup> Floor to 2<sup>nd</sup> floor } in the direction of escalator.

$$m = \frac{20}{10} = 2 \text{ steps/sec}$$

$$\boxed{n = 20 + 10e}$$







#Q: Two cars travel from different locations at constant speeds. To meet each other after starting at the same time, they take 1.5 hours if they travel towards each other, but 10.5 hours if they travel in the same direction. If the speed of the slower car is 60 km/hr, then the initial distance (in km) between the two cars is

$$\frac{D}{140} = 1.5$$

$$\boxed{D = 210 \text{ km}}$$

$$\frac{D}{A + 60} = 1.5$$

$$\frac{D}{A - 60} = 10.5$$

$$D = 1.5A + 90$$

$$D = 10.5A - 630$$

$$0 = -9A + 720$$

$$\boxed{A = 80}$$

#Q: Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all the three cross each other at the same time, is

A 58.8

B 64.4

C 67.2

D 61.6 ✓



$$\frac{54 \cancel{\text{m}}}{(V - 40) \cancel{\text{km/hr}}} = \frac{\cancel{225} \text{ m}}{\frac{50}{2} \cancel{\text{km/hr}}}$$

$$V - 40 = \frac{108}{5}$$

$$V = 40 + 21.6 = 61.6$$

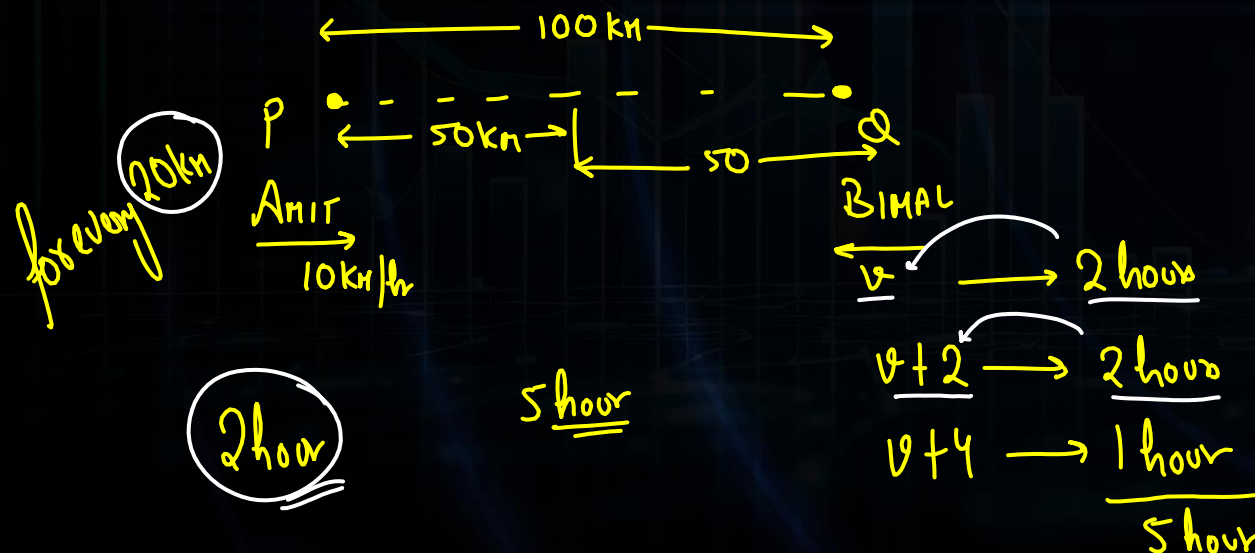
#Q: Two friends, Amit and Bimal, start walking simultaneously towards each other from two points, P and Q, which are 100 km apart. Amit's speed is a constant 10 km/hr. Bimal's initial speed is 'v' km/hr, but for every 20 km that Amit covers, Bimal increases his speed by 2 km/hr. If they meet exactly at the midpoint of the journey, what was Bimal's initial speed, 'v' ?

**A** 10 km/hr

**B** 8.4 km/hr ✓

**C** 8.8 km/hr

**D** 6 km/hr



$$\frac{2v}{T.S} + \frac{2(v+2)}{T.S} + \frac{1(v+4)}{T.S} = 50$$

$$5v = 42$$

$$v = 8.4$$

RAHUL  
LUHAR

#Q: Brishti went on an 8-hour trip in a car. Before the trip, the car had travelled a total of  $x$  km till then, where  $x$  is a whole number and is palindromic, i.e,  $x$  remains unchanged when its digits are reversed. At the end of the trip, the car had travelled a total of 26862 Km till then, this number again being palindromic. If Brishti never drove at more than 110 km/h, then the greatest possible average speed at which she drove during the trip, in Km/h, was

$$\begin{array}{r}
 26861 \\
 25982 \\
 \hline
 26062
 \end{array}
 + \begin{array}{r}
 1 \\
 \text{to} \\
 880 \\
 \hline
 \text{(8 hour Trip)} \\
 800
 \end{array}
 = 26862$$

$$25982 \leftrightarrow 26861$$

$$25982 \longleftrightarrow 26861$$

$$\begin{array}{c}
 \rightarrow \quad \leftarrow \\
 \text{N A M A N} \\
 \rightarrow \quad \leftarrow \\
 \boxed{25982}
 \end{array}$$

$$\begin{array}{r}
 \text{(8 hour Trip)} \\
 26062 + 800 = 26862
 \end{array}$$

$$S_p = 100 \text{ km/hr}$$

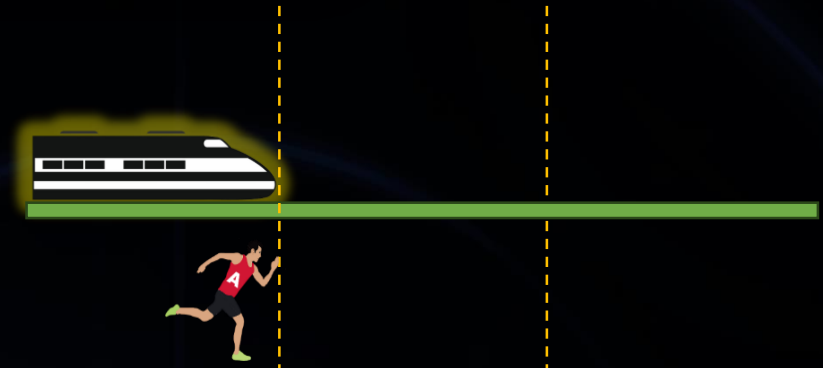
$$26062$$



$$(S_T T) = (L_B + L_T)$$

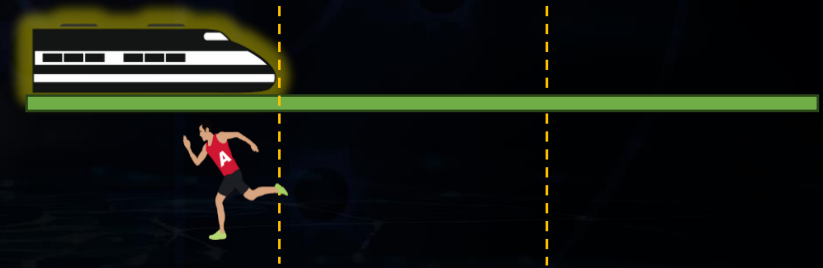


$$S_T T = L_T$$

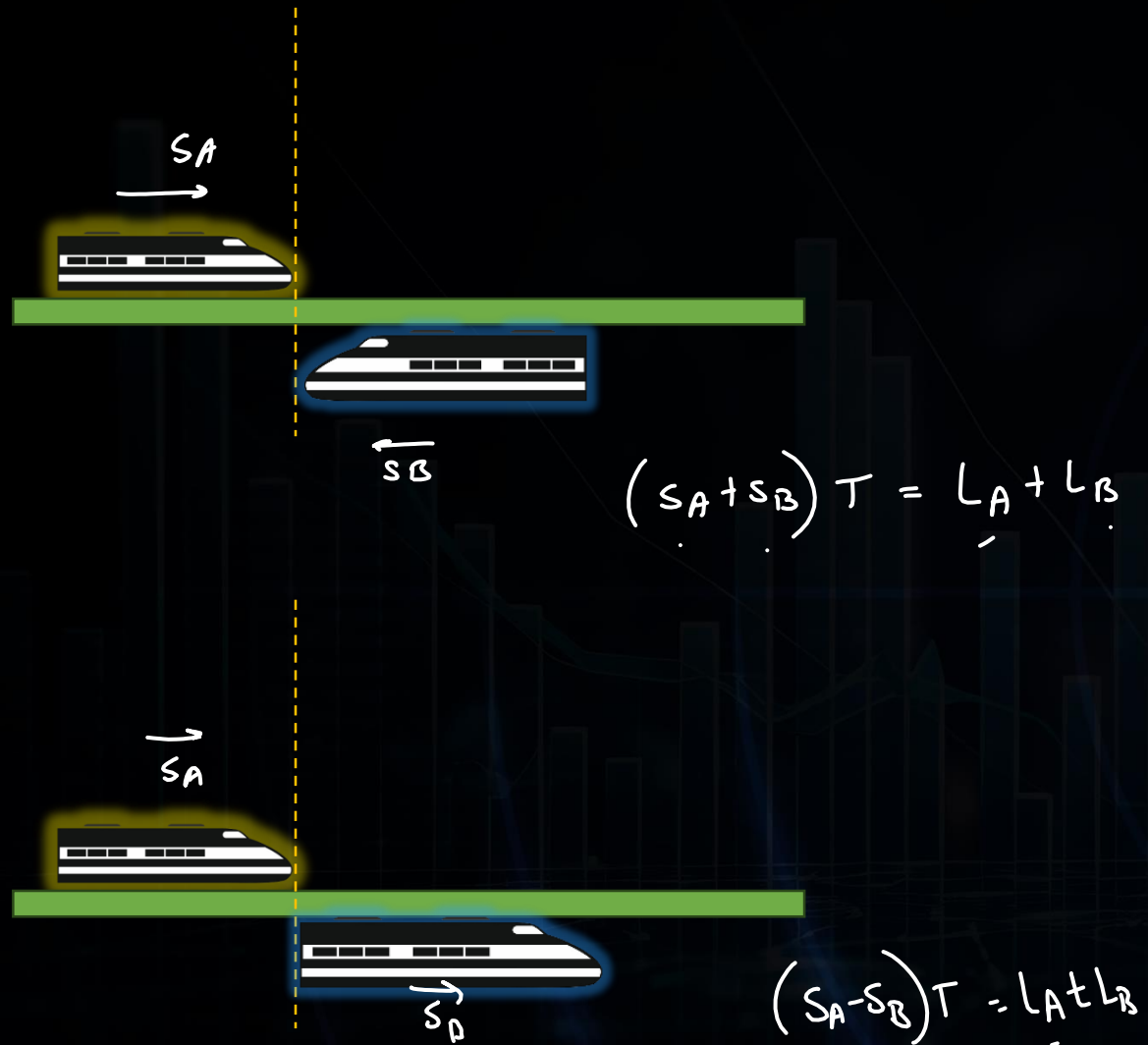


$$(S_T - S_A) T = L_T$$

$$(S_T + S_A) T = L_T$$







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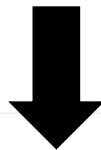
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**THANK YOU**

